

BANARAS HINDU UNIVERSITY

B.Sc. (Hons) Semester IV Examination 2025-26 (NEP)

CHEMISTRY

Paper Code: CHEMJ42 : Organic Chemistry

Ref. No.: Conf/Even/N/2025-26/06

Time: 2½ Hours

Maximum Marks: 70

Instructions: Answer five questions including Question 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Answer the following:

[7 × 2 = 14]

- (a) Write down the cis- and trans-isomers of indigotin and explain their stability.
- (b) Give two suitable methods for the protection of amino (–NH₂) group in organic synthesis.
- (c) Write resonating structures of thiophene using d-orbitals.
- (d) Explain anomeric effect with suitable example.
- (e) Define glycosylation reaction giving suitable example.
- (f) Describe any one method for the synthesis of cyclopropane.
- (g) Complete the following reactions:
 - (i) $\text{Furan} + \text{NH}_3 \xrightarrow[\text{heat}]{\text{Al}_2\text{O}_3, \text{H}_2\text{O}} ?$
 - (ii) $\text{Indole} + \text{HNO}_3, \text{Ac}_2\text{O} \rightarrow ?$

Question 2.

- (a) Give the chemical equations and names of the product formed, when nitrobenzene reacts with— (i) Sn/HCl (ii) Zn/NH₄Cl (iii) Methanolic NaOH and zinc dust (iv) Zn/CH₃OH/NaOH. [4 × 1 = 4]
- (b) Explain the term epimerization using a suitable example. Discuss the interconversion of D-glucose and D-mannose, providing the chemical reactions involved in the process. [4]
- (c) Starting from aniline, how would you prepare 1,3,5-tribromobenzene? Explain why direct bromination of benzene is not suitable for this synthesis. [4]
- (d) An organic compound [A] with molecular formula C₆H₇N on treatment with NaNO₂ and HCl at low temperature give [B]. Compound [B] reacts with cuprous cyanide to give [C] (C₇H₅N). Upon acid hydrolysis, [C] yields a carboxylic acid [D] (C₇H₆O₂). Identify [A] to [D] and write the reactions. [2]

Question 3.

- (a) Describe the Haworth synthesis of naphthalene in detail, including all reagents and steps involved. [5]

- (b) Explain the synthesis of indigotin (indigo) from anthranilic acid. Also describe the vat dyeing process of indigo with suitable reaction. [4]
- (c) Predict the products when naphthalene reacts with— (i) CrO_3 in acetic acid (ii) $\text{Na/C}_2\text{H}_5\text{OH}$ (iii) Conc. H_2SO_4 at 160°C . [3 × 1 = 3]
- (d) Explain why the $\text{C}_1 - \text{C}_2$ bond in naphthalene is shorter than the $\text{C}_2 - \text{C}_3$ bond. [2]

Question 4.

- (a) Write the product(s) in the following reactions: [6 × 1 = 6]
- (i) Anthracene + Maleic anhydride $\xrightarrow{\Delta}$?
- (ii) Anthrone $\xrightarrow{\Delta}$?
- (iii) N-methylformanilide + $\text{POCl}_3 \xrightarrow{\text{heat}}$?
- (iv) Pyridine $\xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) KOH}}$?
- (v) 2-Furaldehyde + $\text{NaOH} \rightarrow$?
- (vi) Pyridine + $\text{CH}_3\text{CO}_3\text{H} \rightarrow$?
- (b) Compare and discuss the aromatic character of furan, thiophene and pyrrole. Arrange them in the decreasing order of aromaticity. [4]
- (c) Draw all the conformations of 1,2-dimethylcyclohexane. Identify which isomer is more stable and why. [4]

Question 5.

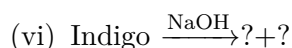
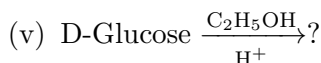
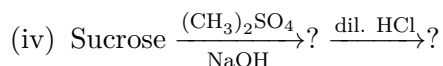
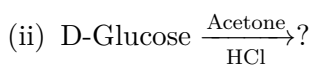
- (a) Discuss the mechanism of any three of the following: [3 × 2 = 6]
- (i) Chichibabin reaction
- (ii) Feist-Bénary synthesis
- (iii) Skraup synthesis of quinoline
- (iv) Fischer indole synthesis
- (b) Describe the reaction and mechanism for synthesis of fluorescein dye. [4]
- (c) Write all the steps involved in the synthesis of osazone from D-glucose. Explain why glucose, fructose and mannose form the identical osazone with excess phenylhydrazine. [4]

Question 6.

- (a) Discuss Baeyer's strain theory for the stability of cycloalkanes. Calculate the 'Angle strain' for cyclopropane and cyclobutane. Why does this theory fail to explain the stability of higher cycloalkanes? [6]
- (b) Draw the energy profile diagram for different conformations of cyclohexane. Arrange them in the decreasing order of stability and give reasons for the same. [4]
- (c) Describe the use of silyl ethers in protecting alcohols. How are these groups introduced and removed? [4]

Question 7.

(a) Complete the following chemical equations with appropriate structures: $[6 \times 1 = 6]$



(b) How will you prepare the following from benzene diazonium chloride? $[3]$

(i) Biphenyl

(ii) Phenylhydrazine

(iii) Azobenzene

(c) Compare the structures of starch and cellulose. Explain the difference between α -glycosidic and β -glycosidic linkages and how they affect the overall shape of the molecule. $[3]$

(d) Describe the structure of sucrose. Why is it called non-reducing sugar? $[2]$

Question 8.

(a) Why is protection of functional groups necessary in organic synthesis? Mention the three essential criteria for an ideal protecting group. $[4]$

(b) How will you reduce a keto group in a molecule that also contains an aldehyde group without affecting the aldehyde group? $[3]$

(c) Explain the protection and deprotection of the following functional groups with suitable reagent: $[3]$

(i) Carboxyl group

(ii) Terminal triple bond

(iii) 1,2-diol

(d) Compare the basic strength of pyridine and pyrrole. $[2]$

(e) Write the chemical equation and discuss the reaction between furan and maleic anhydride. $[2]$
